

CAMFIL APC - AIR POLLUTION CONTROL

CLEAN AIR SOLUTIONS



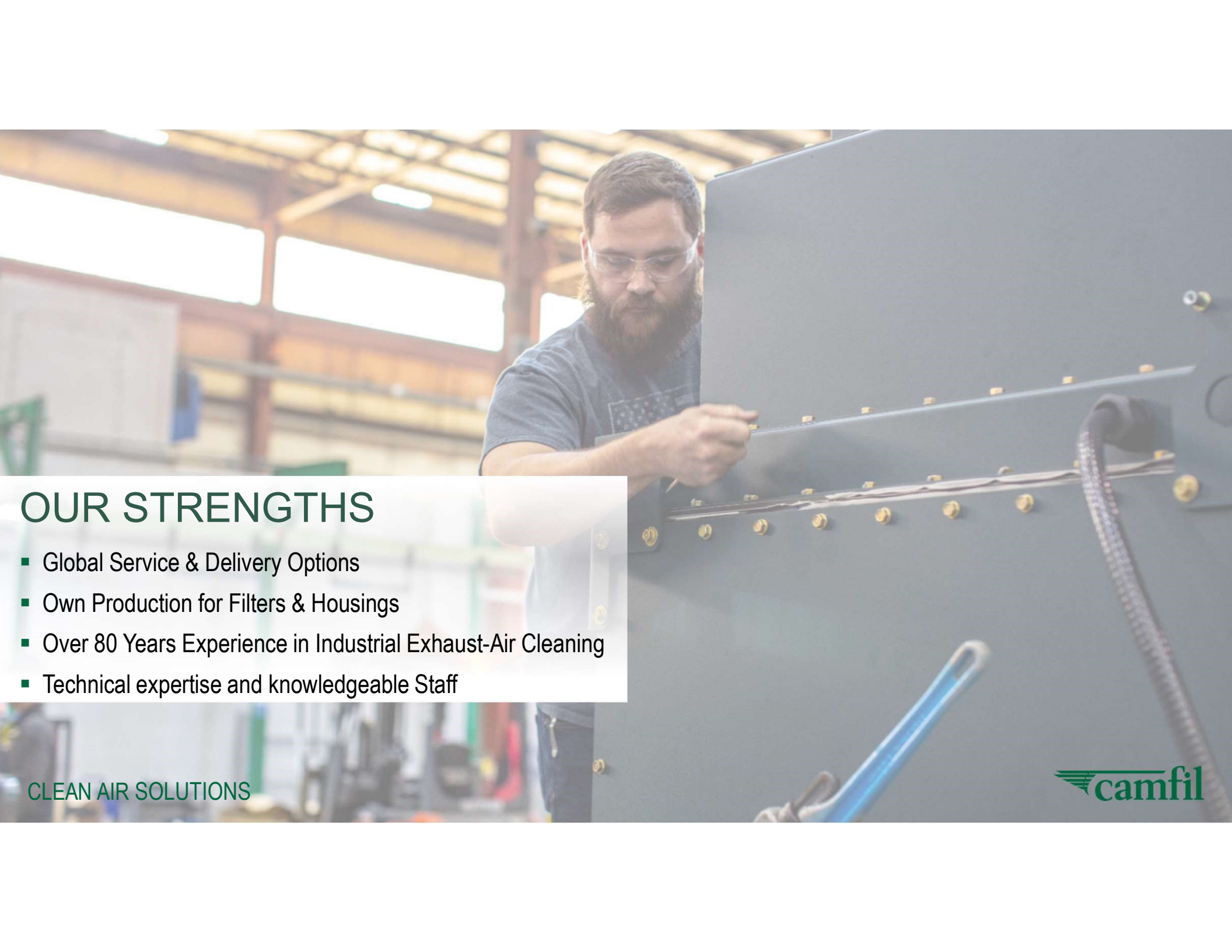


WHY DO WE DO IT ?

- We firmly believe that **clean air is a fundamental human right**. That's why we develop and manufacture innovative solutions for industrial exhaust-air purification.
- Every manufacturing process creates some kind of dust, mist or fume. These byproducts can be harmful to workers, and they can cause machinery to become less efficient.
- Camfil APC's product range is designed to **keep your workforce safe and healthy, and ensure that your operations and processes remain as efficient as possible**.

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OUR STRENGTHS

- Global Service & Delivery Options
- Own Production for Filters & Housings
- Over 80 Years Experience in Industrial Exhaust-Air Cleaning
- Technical expertise and knowledgeable Staff

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COMPREHENSIVE PRODUCT PORTFOLIO

- We offer unique industry expertise, customer intimacy and customized air filter solutions for most production areas.
- From a simple filter system to complex turnkey solutions, from the initial design of the plant to the final completion.

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A photograph of two men in a factory or industrial setting. The man on the left is younger, with short brown hair and glasses, wearing a dark blue shirt with an orange collar. The man on the right is older, with grey hair and glasses, wearing a dark blue shirt with an orange collar and a 'camfil' logo on the chest. They are both smiling and looking towards each other. The background is blurred, showing industrial equipment and green structural elements.

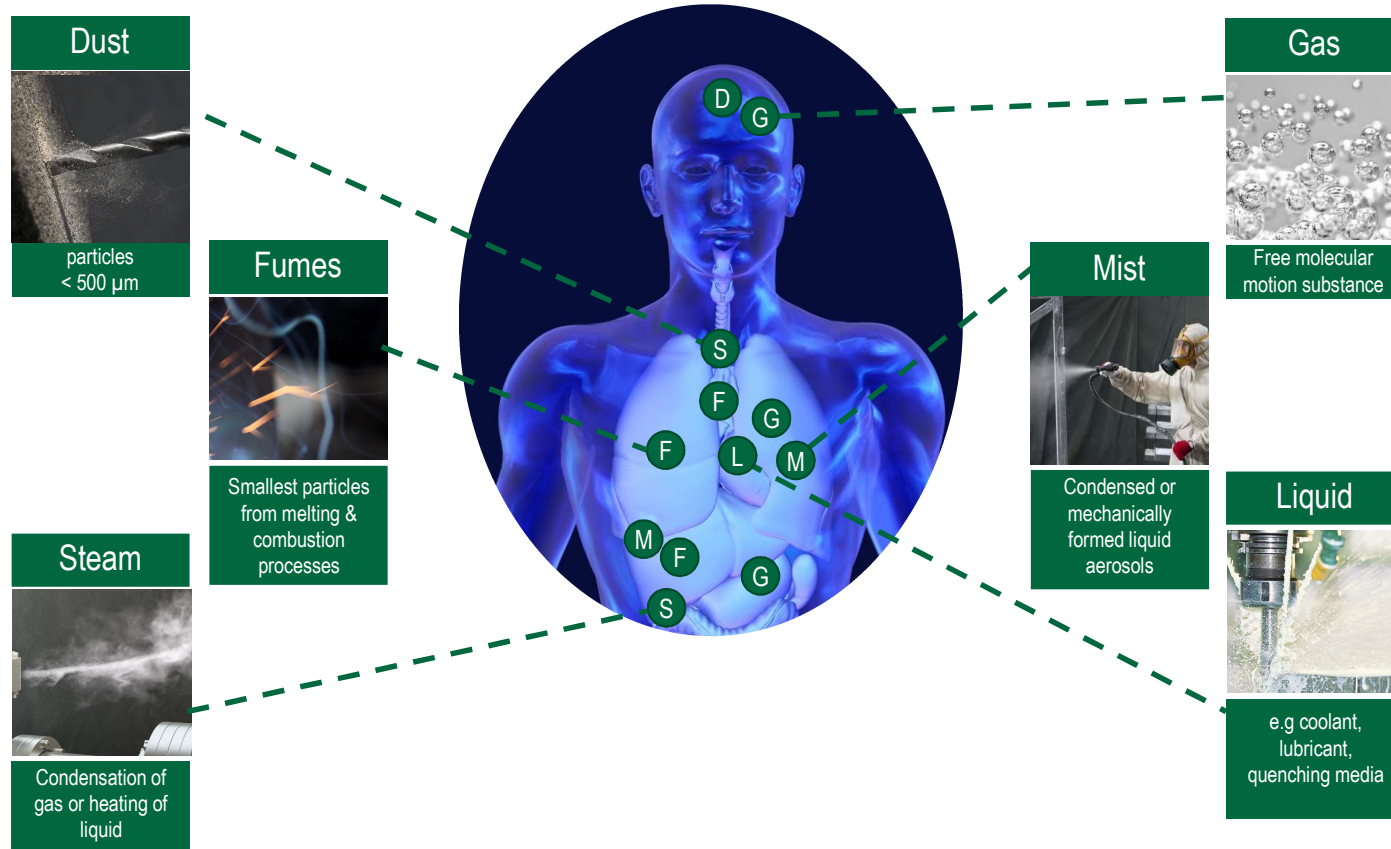
TRUST OUR EXPERTS

Our engineers and employees are absolute professionals with many years of experience in the development, production and technical support of air pollution control systems.

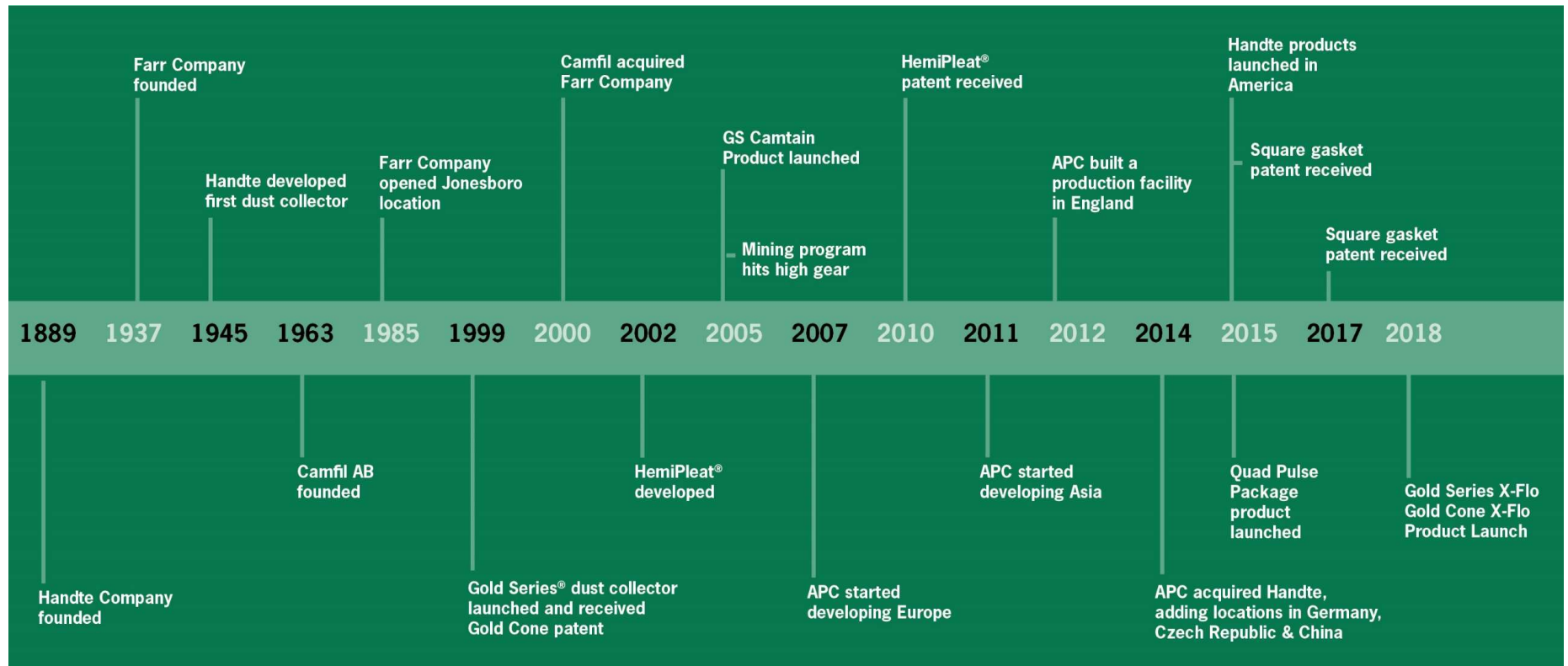
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INDUSTRIAL POLLUTANTS AND INTERACTION ON HUMANS



HISTORY OF CAMFIL APC



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CAMFIL APC PRODUCTION FACILITIES

Production Jonesboro, AR, USA



Production Tuttlingen,
Germany



Production Chomutov,
Czech Republic



Production Taicang, China



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KEY MARKETS



Metal OEMs

Metal Processing

Automotive

Pharmaceutical

Mining

Food & Beverage

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AUTOMOTIVE



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MEDICAL



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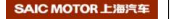
AEROSPACE



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CUSTOMER REFERENCES

 Audi	 BMW	 Mercedes-Benz	 Volkswagen	 Ford	 GM	 TESLA	 FIAT	 HONDA	 TOYOTA
 RENAULT Passion for life	 一汽-大众 FAW - VOLKSWAGEN	 SAIC MOTOR 上海汽车	 MICHELIN	 härning VÄRRE FÖR PÅSEDDRETTAR	 ZF	 Continental	 LUK INA FAG SCHAEFFLER GRUPPE	 SACHS	
 COMAU	 FANUC	 VOITH	 WACKER	 TRUMPF	 oerlikon	 AMADA	 GROB	 chiron	 HERMLE Kitchen Whirls
 Bystronic	 EMAG	 Mazak	 MESSER Cutting Systems	 BASF We create chemistry	 Schindler	 BOSCH	 +GF+	 WÜRTH	 STORZ KARL STORZ – ENDOSKOP
 PLIVA	 Roche	 TEVA	 Nestlé	 DOW	 Abbott	 SANOFI	 gsk	 Pfizer	 syngenta
 Lilly	 CORTEVA agriscience	 GOWAN MILLING	 amneal	 hikma.	 PIONEER	 Takeda	 Wyeth	 MERCK	 AstraZeneca
 Janssen	 Lundbeck	 MSD	 AESCULAP	 NOVARTIS	 P&G	 SCA	 SANDOZ A Novartis Division	 BAYER	 MONSANTO
 IMA Sustain Ability	 FREEPORT-McMoRAN	 ThyssenKrupp	 BALDWIN RICHARDSON FOODS CO.	 Maruchan	 Russel Metals	 JPLW COMPANIES	 3M	 AIRBUS	 MTU

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PRODUCT PORTFOLIO

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APPLICATION CATEGORIES

DRY FILTRATION



WET FILTRATION



OIL & EMULSION



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DRY FILTRATION – PRODUCTS AT A GLANCE

GOLD SERIES® X-FLO



QUANTUM SERIES®



QUAD PULSE PACKAGE



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GOLD SERIES X-FLO®

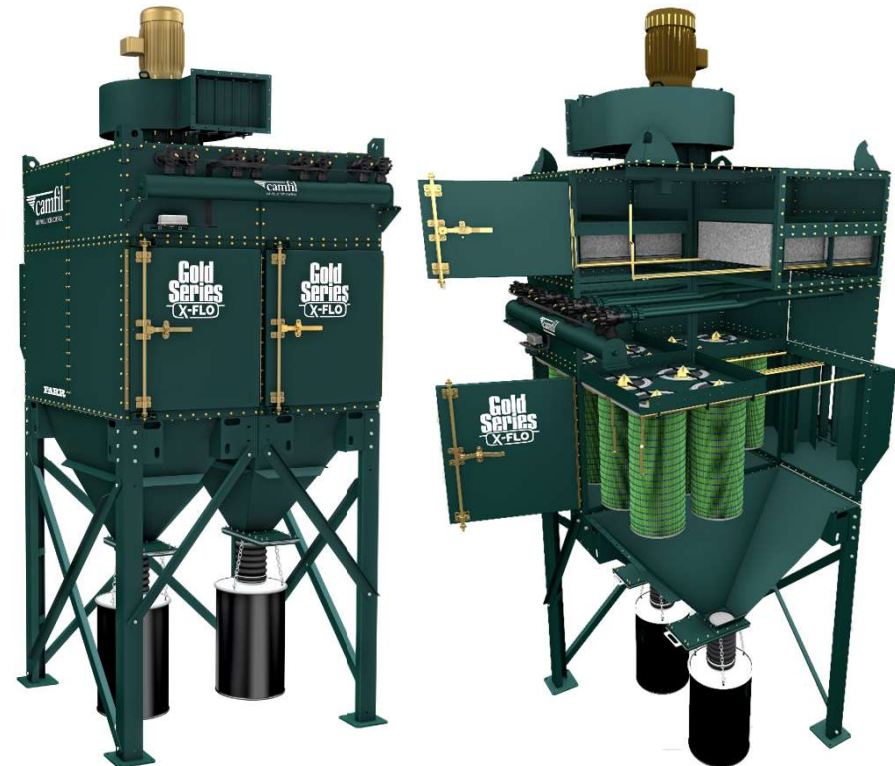
Handles all kinds of toxic and combustible dusts, smoke and fumes, including fine, fibrous and heavy dust loads.

Applications:

- Metalworking, pharmaceutical, food & beverage, chemical processing, mining

Specifications/Benefits:

- ✓ Modular design for different sizes & configurations
- ✓ Airflows up to 6,000 cfm per module
- ✓ Four cartridges per module
- ✓ More filter media, same footprint
- ✓ Maximum durability
- ✓ Easy to use & maintain



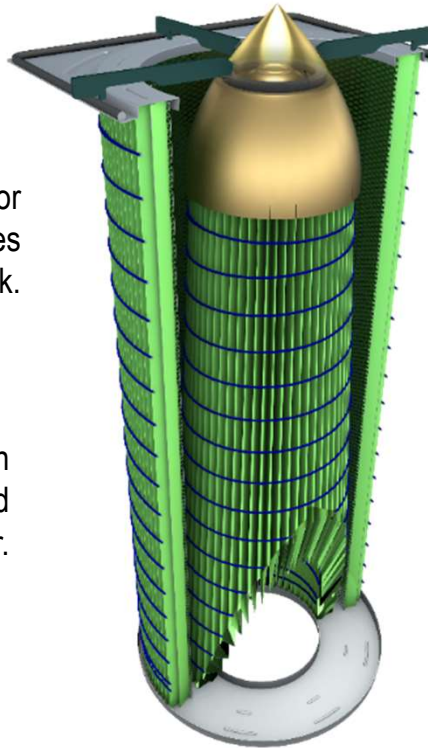
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GOLD CONE X-FLO

Cone technology Patent allows for taller and larger cone that initiates cleaning higher on the outer pack.

More total media, with 50% more hemi-pleated media than GC filter.



Optimized open area around cone tip, lowers initial pressure drop by 30% extending filter life.

GCX Cone has over 50% more effective “downward facing” media, sending more dust out of the collector with every pulse.



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GOLDLINK: A CONNECTED PRESSURE DROP SENSOR

Phase 1

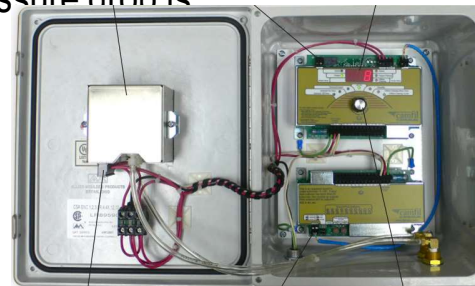
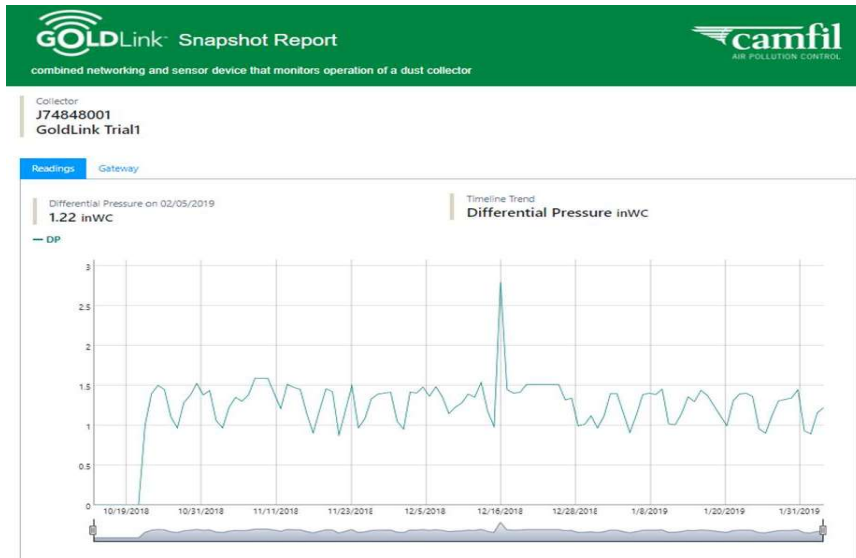
- Develop software
- Develop hardware to be installed in control panels of each collector

Phase 2

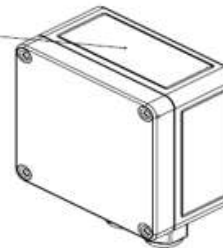
- Install all units Q1 2019
- Next step in customer service begins – We tell the customer what their pressure drop is

Phase 3

- Design new timer board and controls to fully incorporate IoT into panel Q3 2019



3G CELLULAR STRIP
ANTENNA UNDER
THIS SURFACE



Welding



GS48 - Nissan



GS48 - Ambient

Thermal Spray



GS48 - Arkivator



GS20, GS24 & 2xGS32 - RRBS

Food



GS08 - KP Snack Foods



GS24 - Crown Bevcan s.r.o.

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Chemical



2xGS20 - Unilever



2xGS32 – Carbon black

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Mining



Phelps Dodge - Copper Mine



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QUANTUM SERIES®

The Quantum Series® is a compact and powerful, fine dust and fume collector designed for the metal fabrication, thermal cutting and welding markets.

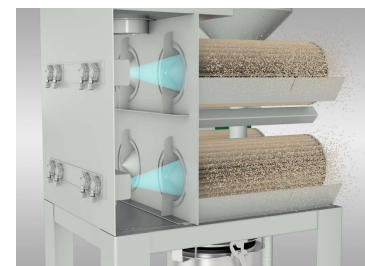
Applications:

- Welding (laser/manual), laser & plasma cutting, none-explosive fine dust

Specifications/Benefits:

- ✓ Dual pulse technology
- ✓ Integrated spark arrestor
- ✓ Plug & play installation
- ✓ Maximum reliability
- ✓ Minimum footprint
- ✓ High efficiency PTFE filter cartridges
- ✓ Three different sizes available – up to 5,500 m³/h

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QUANTUM SERIES - MAIN APPLICATIONS

Manual Welding



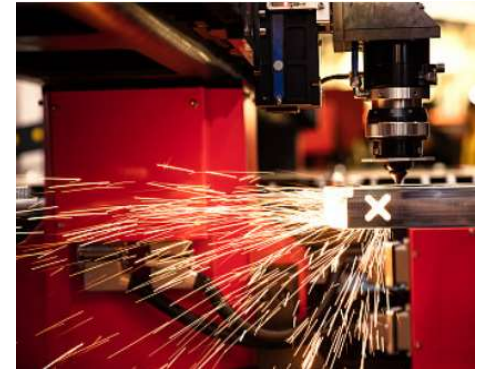
Laser Welding



Plasma Cutting



Laser cutting



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FEATURES AND BENEFITS

QUANTUM
SERIES

Safe

- Integrated cyclone spark arrestor
- Economic fire suffocation package

Clean

- Powerful pulse cleaning system
- PTFE filter cartridges
- Separately dust collection bin

Compact

- Plug and play
- Small footprint

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MODULAR DESIGN

QUANTUM
SERIES

- Each model can be configured to meet the specific challenges faced by the customer.
- All models/capacities have use same floor space
- Quantum is manufactured in China and in Europe
- ePTFE cartridges
- IP protected



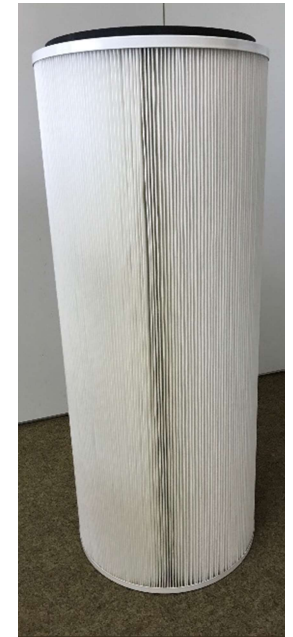
QS2



QS4



QS6

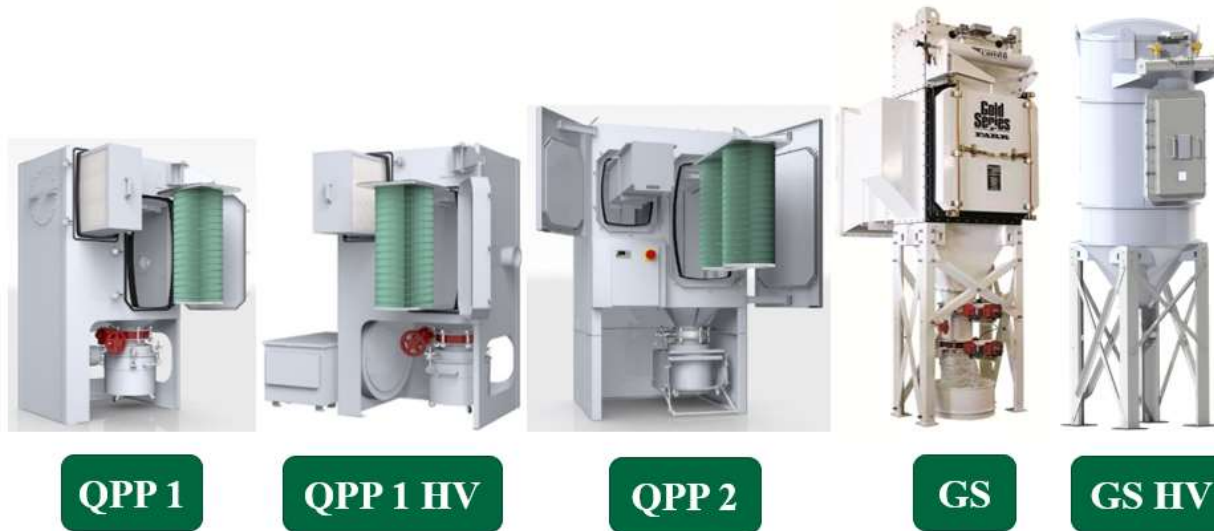


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BIO PHARMA PRODUCT OFFERING

- GS CamTain is the multi-purpose solution for Pharma
- QPP1 & 2 are the compact units required to optimize footprint
- Synergies within Camfil Group allow to offer complete solutions for OSD applications



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GOLD SERIES® CAMTAIN

The dust collector designed specifically for pharmaceutical and containment applications.

Applications:

- Tablet presses + coating, fluid bed drying, spray drying, blending, granulation, general room ventilation

Specifications/Benefits:

- ✓ Safe-change containment systems for filter cartridges & discharge system
- ✓ High-efficiency filters up to MERV 16 stop 99.99% of the dust at 0.5 microns
- ✓ Specially treated filter media repels fine particulates for a lower pressure drop and longer filter life



QUAD PULSE PACKAGE SERIES

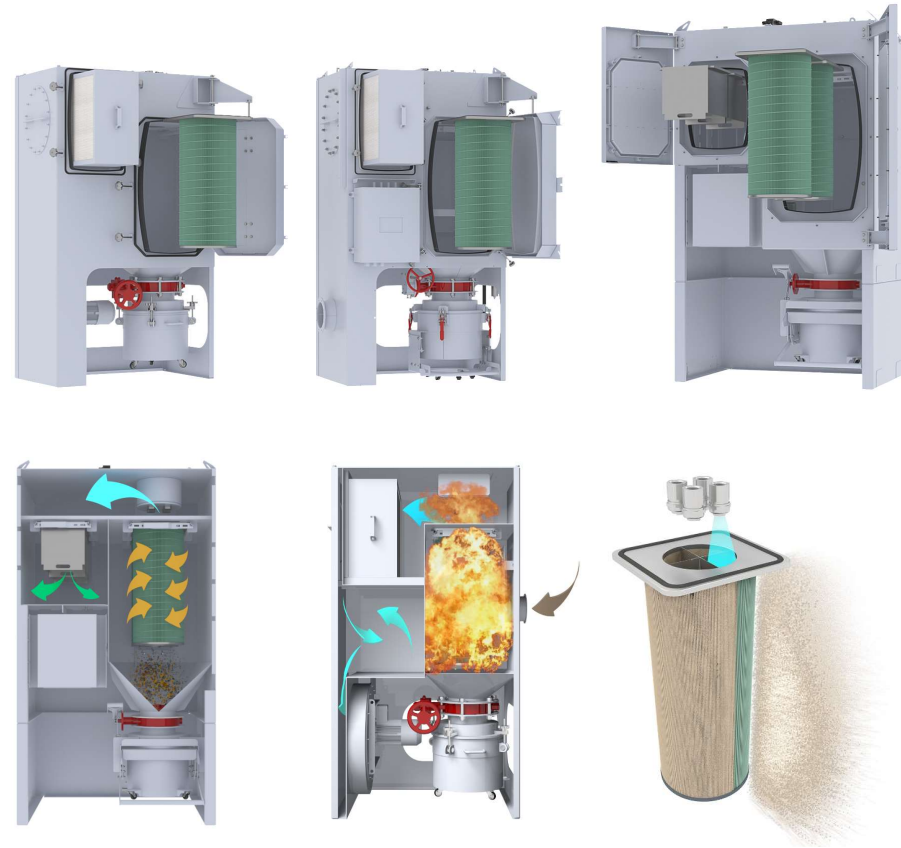
A product family with a mission: cleanliness, safety and efficiency for hazardous dust.

Applications:

- Tableting, filling, coating, granulating, drying, mixing, blending, packaging, central vacuum

Specifications/Benefits:

- ✓ Compact design / 1 or 2 filter cartridges
- ✓ Clean online without interrupting operations
- ✓ Easy installation
- ✓ BIBO safe-change
- ✓ Integrated explosion protection
- ✓ ATEX certified
- ✓ HV-applications



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The logo features a stylized green wing icon to the left of the word 'camfil' in a green, lowercase, sans-serif font.

APPLICATION CATEGORIES

DRY FILTRATION



WET FILTRATION



OIL & EMULSION



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HANDTE WET SCRUBBERS



The systems with the most natural „filter“ media ever: **Water!**

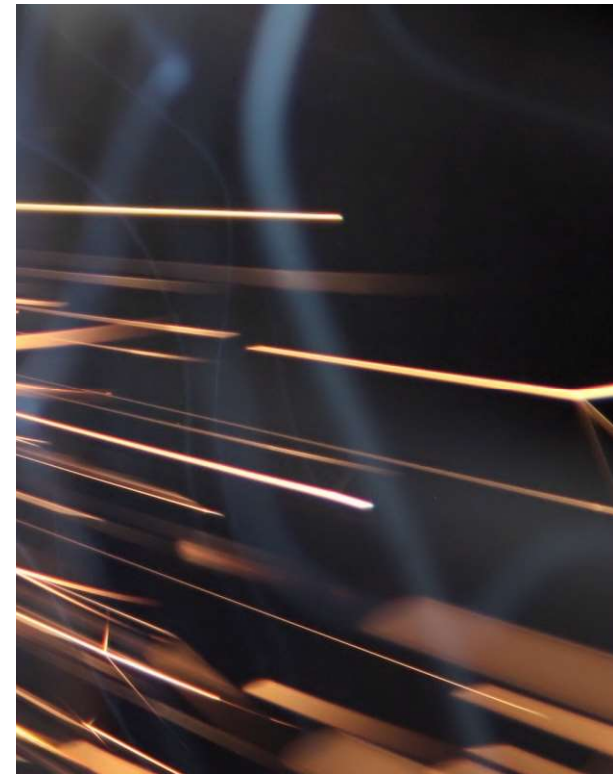
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The logo for Camfil, featuring a stylized green wing-like graphic to the left of the word "camfil" in a green, sans-serif font.

HANDTE WET SCRUBBERS - AN ALTERNATIVE TO DRY FILTERS

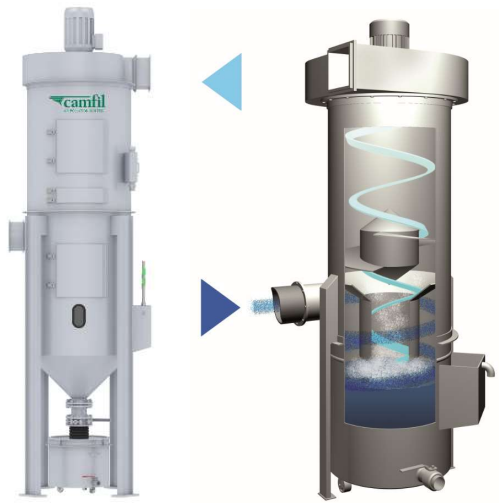
Wet scrubbers are good problem solver for:

- Filtration sticky dusts & fibers (e.g. polishing with paste)
- Processes with a high fire risk and sparks (e.g. grinding steel)
- Filtration of explosive dusts (e.g. aluminum, magnesium, titanium)
- Combined processes with dry and wet processing cycles
- Filtration of emission mixtures out of dust, gas, aerosols, vapors, fumes



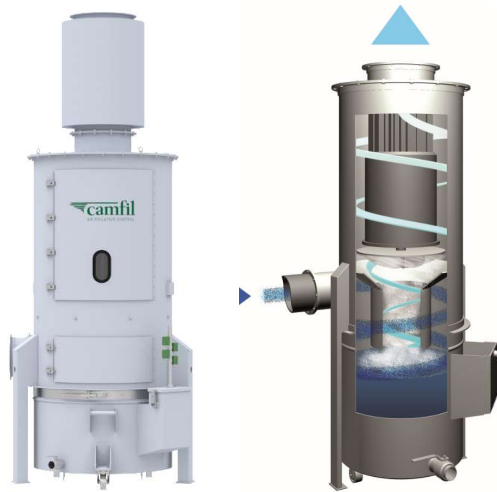
WET FILTRATION – PRODUCTS AT A GLANCE

VORTEX

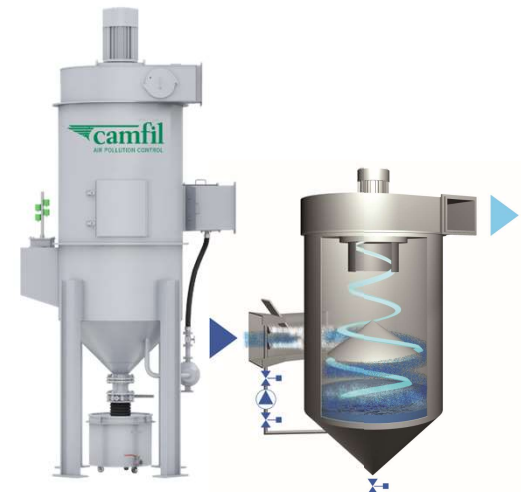


Vortex scrubber for sticky fibres and lint.

VORTEX DUAL



VENTURI



High performance wet scrubber for higher separation efficiency.

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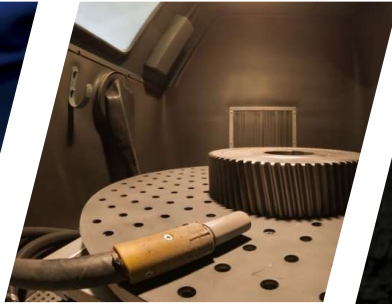


TYPICAL MARKETS – WET FILTRATION

Grinding



Blasting



Drilling



Brushing



Fettling



3M

 **Russel Metals**

JFW C O M P A N I E S

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APPLICATION CATEGORIES

DRY FILTRATION



WET FILTRATION



OIL & EMULSION



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Wurth-Arnold in Shenyang, China: Turnkey Project



Before

Nov 2017



After

Jan 2018

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MIST FILTRATION – PRODUCTS AT A GLANCE

HANDTE OIL EXPERT



HANDTE EM PROFI



HANDTE EM EXPERT



HANDTE EM-O COMPACT



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OIL EXPERT

The oil mist collector for extremely efficient separation of ultra-fine mists and fumes.

Applications:

- Oil mist separation, oil mist collection, oil mist filtration

Specifications/Benefits:

- ✓ Progressive filtration design for highest separation efficiency
- ✓ Modular design for flexible configuration
- ✓ Innovative new filter media for a long lifetime
- ✓ Very easy filter change for simple, clean maintenance
- ✓ Collection efficiency at 99,95% on 0,3 micron and higher particle size
- ✓ Suitable for continuous 3-shift operation
- ✓ Plug & Play



EM PROFI

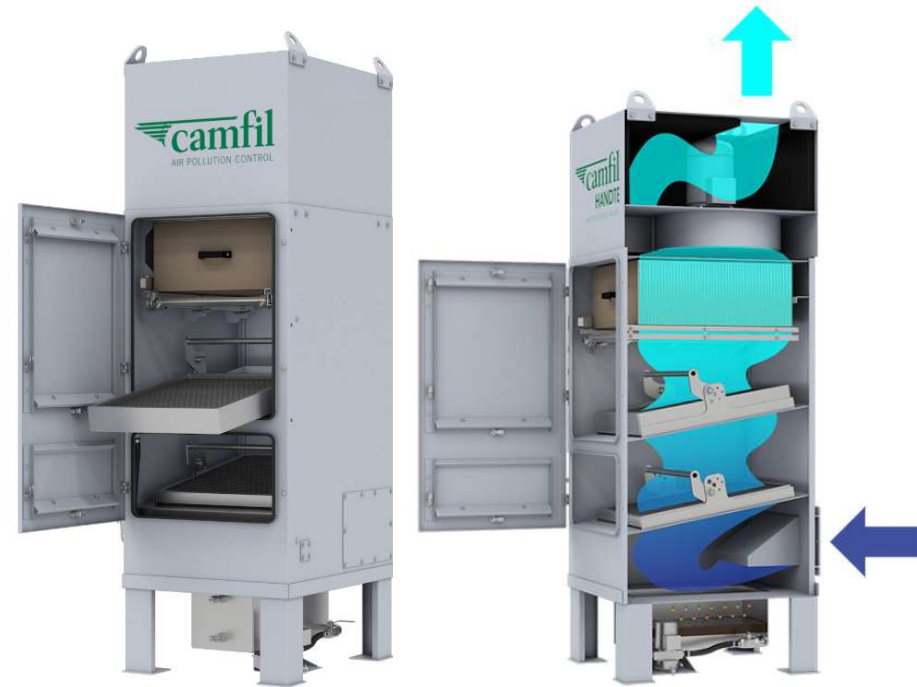
Collects coolant mist from low speed/low pressure machining processes in shops and manufacturing environments.

Applications:

- Turning, milling, honing

Specifications/Benefits:

- ✓ Collection efficiencies up to 99,95%
- ✓ Self-cleaning
- ✓ Automatic sprayer system reduces filter buildup
- ✓ Corrosion-proof and washable two-stage filtration system
- ✓ Easy maintenance
- ✓ Modular assembly



EM EXPERT

The coolant mist collector for extremely efficient separation of ultra-fine cooling lubricant and fumes.

Applications:

- Grinding, cutting, drilling

Specifications/Benefits:

- ✓ Highly efficient, multi-stage filtration
- ✓ Flexible and modular design
- ✓ Multiple units can be combined for any size application
- ✓ Self-cleaning filtration media
- ✓ Long filter life with minimal maintenance requirement
- ✓ Optional HEPA filter
- ✓ Suitable for continuous 3-shift operation



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EM-O COMPACT

The combined emulsion / oil separator was specially developed for simple and flexible integration in machine tools.

Applications:

- Full lubrication applications with emulsion or oil and for selected minimum lubrication applications with air volume from 800 - 1,200 m³/h

Specifications/Benefits:

- ✓ Compact construction
- ✓ Maximum filter area
- ✓ >F9 separation efficiently (filtration) with optional HEPA filter
- ✓ Long service life
- ✓ Energy-efficient
- ✓ Plug & work

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APPLICATION EXAMPLES



For Cleaner Production

Standard & Customized Solutions from Camfil APC :

- ✓ Compact Construction
- ✓ Clean Design
- ✓ Small Footprint
- ✓ Different Air Flows



Tailored for your needs!



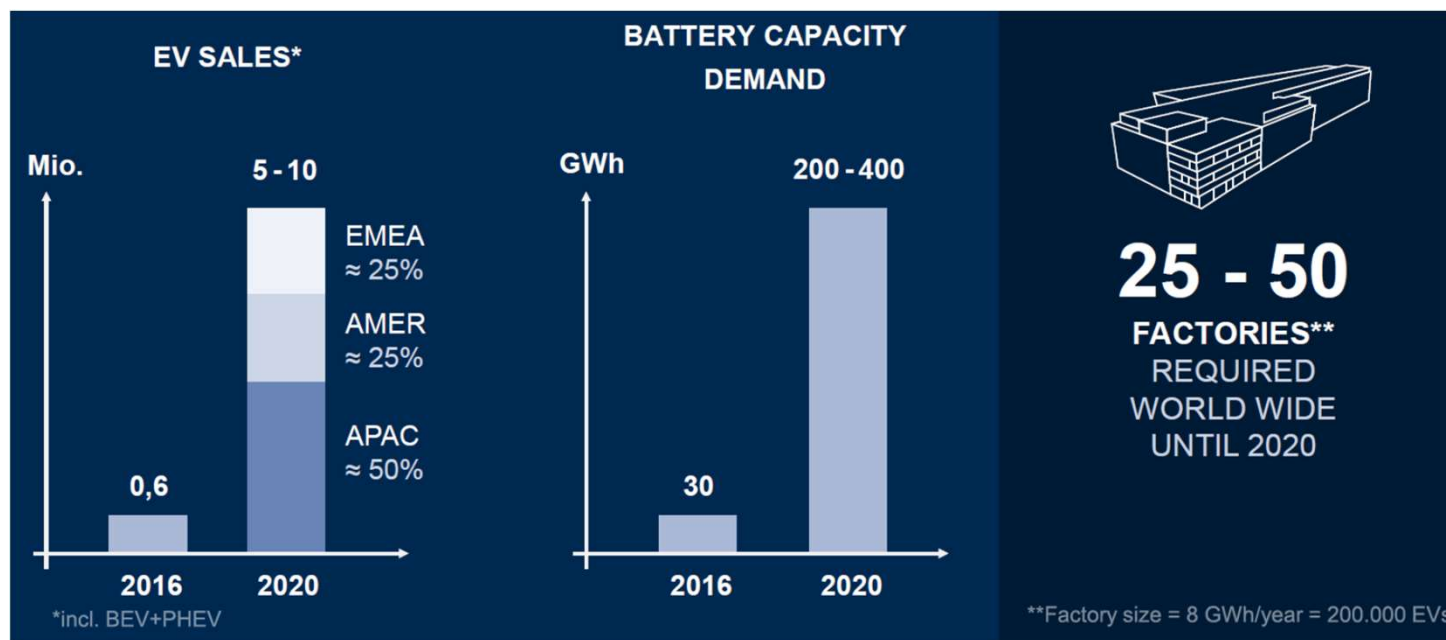
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E-MOBILITY

E-MOBILITY: BATTERIES AND ELECTRIC ENGINES THE TURNING POINT IN THE HISTORY OF ENGINE AND TRANSMISSION

In 2030 in Asia > 50% of the cars will have an electric engine



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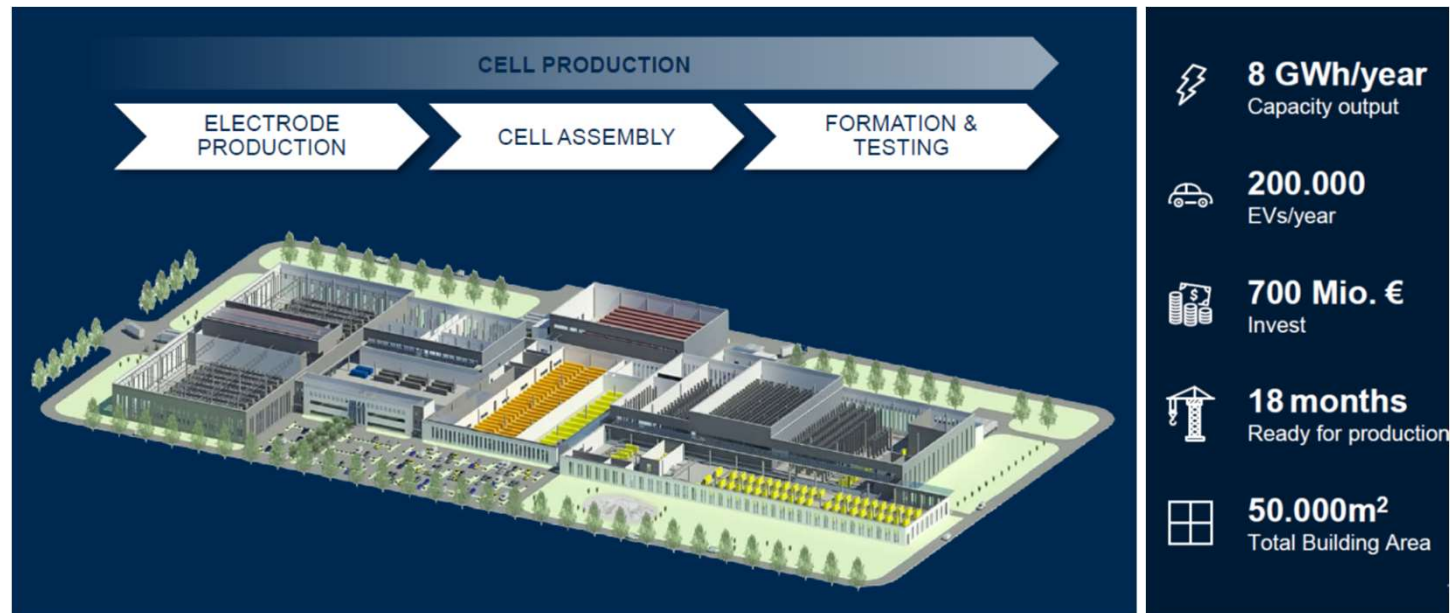
Welding Dry dust

Molecular solutions to capture chemical gases



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E-MOBILITY: BATTERIES



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E-MOBILITY: ELECTRIC ENGINES

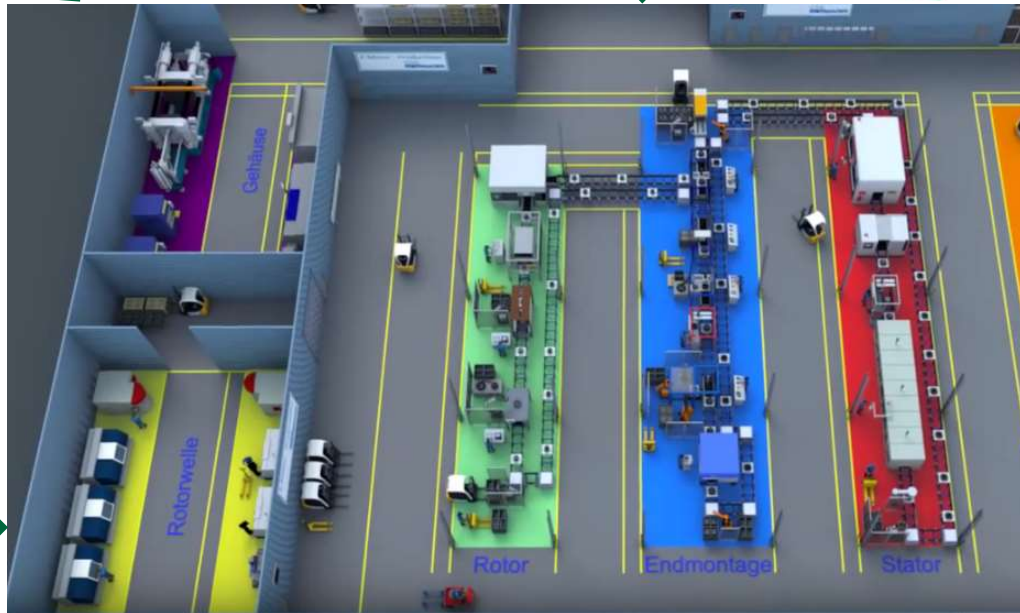


diecast Alu housing



Shaft machining

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Welding



Typical view of "P" type unit with pre- and after-filters

Impregnation

Toxic Dust



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Imprägnierzellen

Abkühlstrecke

Ofen

Ofenlinie

110000 75 90 100 110 120 130 140 150 160 170 180 190 200 210 220 230 240 250 260 270 280 290 300 310 320 330 340 350 360 370 380 390 400 410 420 430 440 450 460 470 480 490 500 510 520 530 540 550 560 570 580 590 600 610 620 630 640 650 660 670 680 690 700 710 720 730 740 750 760 770 780 790 800 810 820 830 840 850 860 870 880 890 900 910 920 930 940 950 960 970 980 990 1000



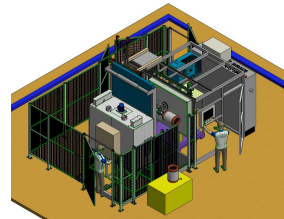
CAMFIL IS A GLOBAL PARTNER FOR E-MOBILITY PROJECTS



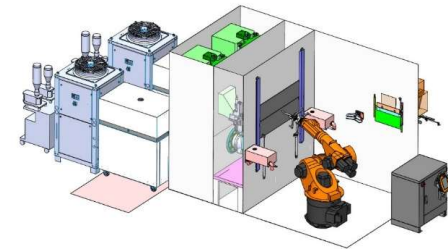
Audi; Block-Cleaning,
Quantum HV



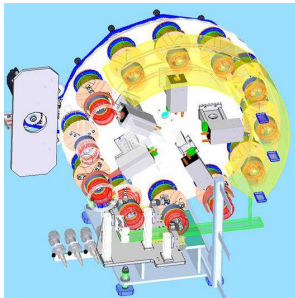
Grob; Hairpin Technology,
Quantum



Grob; Impregnation
Molecular



Chiron; Laser-cladding,
Quantum HV



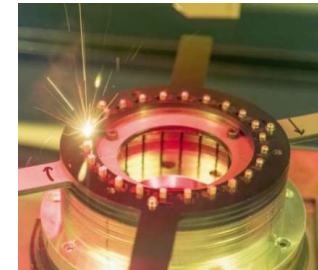
Felsomat; Dripping,
Impregnation Molecular



BMW, Engine Assembly
Camcleaner



VW; Laser de-
Impregnation
MF Compact-unit



Gehring; Hairpin-
Technology, Quantum

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THANK YOU
FOR YOUR ATTENTION

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